Spatial Grounding and Mathematical-Focus Containment in View-Conditioned VR Tutoring

Supplementary Material: Prompts, Visibility Snapshots, and Annotation Materials

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This document supports the paper on view-conditioned hint generation in MathBuild VR and its tutoring component, ViewTutor. It specifies the exact prompts, the visibility-snapshot format, the evaluation set, the annotation criteria, and additional breakdowns. Companion files (prompts, JSON examples, annotated data) are in the accompanying package. All hints were generated from *scripted* snapshots; no learner data are involved.

S1. Pipeline and deterministic verification

When help is requested, the VR client sends the question identifier, the learner's construction state, and a region-level visibility snapshot to the backend. A deterministic evaluator, independent of the LLM, compares the construction with the verified solution structure of the task type and returns a *construction status* (correct, incomplete, or incorrect) with a list of verified errors. Correct constructions receive a fixed confirmation and no LLM call. For incomplete or incorrect constructions, the prompt payload contains the story and question, the stated quantities, the verified arrangement, the status, the verified errors, the allowed numbers, and (view-conditioned condition only) the visibility context. A rule-based answer-protection layer screens the generated text; a blocked hint is replaced by a checked fallback. In the reported run, no hint was blocked.

Relevant-region selection (evaluation only). The prompt never receives a relevant-region flag; the model decides which region the learner needs. For the scripted views, one region per state is designated relevant by a fixed, task-specific priority order over discrepancies between the construction and the story:

Task type	Priority order of discrepancies (first match = primary)
Equal groups	group larger than the story's group size; extra group beyond the story's group count; group smaller than the group size
Equal sharing	recipient share larger than the story's share; recipient share smaller than the story's share
Join	first part differs from the story; second part differs (or, with a single destination, the destination total differs)
Remove	removed collection differs from the story; start/remaining collection differs
Part-to-part ratio	first part differs; second part differs

States without an identifiable discrepant region were not eligible.

S2. Visibility-snapshot format

Request-side snapshot. For each workspace region: `region_id`; `visible_fraction` $f \in [0, 1]$; `label_visible`; `label_seen_this_attempt`. In the headset client, f is the share of nine sample points on the region's top face (corners, edge midpoints, centre) that lie inside the headset viewport and are not occluded (ray-cast test); label visibility is computed separately from f, and a label counts as visible when its own visible fraction is at least 0.5. In the scripted evaluation the values were assigned directly (below) in the same format.

Prompt-side context. The model receives, per region, a child-facing `region_name`, a visibility class, `label_visible_now`, `label_seen_this_attempt`, and `learner_has_seen_name`. It receives no imagery, coordinates, or camera pose. Classes: *not visible* ($f < 0.10$), *partially visible* ($0.10 \leq f < 0.70$), *mostly visible* ($f \geq 0.70$).

Scripted views. f refers to the relevant region unless stated.

View	Relevant region	Other regions
all-visible	$f = 1$	$f = 1$
relevant-only	$f = 1$	$f = 0$ (labels previously seen)
partial	$f = 4/9 \approx 0.44$	$f = 1$
hidden-seen	$f = 0$, label previously seen	$f = 1$
hidden-unseen	$f = 0$, label never seen (synthetic stress test)	$f = 1$

In the scripted views, `label_visible` was set to true whenever $f > 0$ (the headset rule above is not simulated). Serialized examples for one state under all five views are in `examples/visibility_snapshots.json`.

S3. Prompts

Both conditions use the same 15 shared rules, the same strict output schema, and the same per-request payload, except that the reference condition omits the visibility context and the viewpoint guidance. The instruction texts below are exact and identical across all requests within a condition; only the payload varies. One representative payload is shown; every payload is built by the same code from the question, the verified status, and the view. The listings below are the exact instruction texts (typographic quotes normalized for printing).

S3.1 Shared instructions (both conditions)

```
You are a math tutor for children aged 6 to 8, inside a virtual-reality activity where they build word
    problems with objects.
Python has already checked the learner's work. Use only the verified facts you are given. Never decide
    yourself whether the work is correct.
The learner asked for: Can I get a hint?. Goal: Give exactly one small, useful hint based on the current
    arrangement without giving the answer or completing the task for the learner.

Rules:
1. Never reveal the final answer, and never give an operation that directly produces it.
2. Never invent learner actions, written answers, spoken claims, objects, people or relationships.
3. Use only numbers listed in allowed_numbers. Prefer no numbers.
4. Be brief (at most 45 words), kind, and grounded in the visible arrangement. Use the object names given.
5. Story help explains what the story means; check-work asks the learner to inspect one part of the
    arrangement (do not tell them what to move); a hint gives exactly one small step.
6. If the learner wrote a message, answer it and stay on this problem. Do not tell stories, poems or jokes.
7. Do not repeat the wording of recent tutor messages.
8. arrangement_description says what is already on the desk. Never tell the learner to place, add, make or
    create anything it says is already there or already complete.
9. The collections, groups and recipient areas it lists already exist. Ask for a new group only when it says
    there are fewer groups than the story.
10. Never put a region name in quotation marks and never use generic labels such as "the objects to share" or
    "the objects to try". Use the story's own people, object names and the labels the learner sees.
11. Prefer a gentle noticing prompt over a direct instruction. Do not tell the learner how many objects to
    move or add; point to the part of the story to match and let the learner decide. Every hint must contain
    exactly one clear action word the learner can do, such as look, check, compare, count or notice, then
    may end with a short question (for example "Check what the story says about Marcie's part. Does your
    collection match?"). Never reply with only a question and no action word, and do not use first, then or
    next.
12. Do not combine a place with the number the story needs (for example "Group 4 ... the story needs 3
    groups").
13. Do not ask the learner to find or pick an object they cannot see; ask what the story says about that part
    instead.
14. Write at most two short sentences. If a turn phrase is required it is the first sentence, and the second
    sentence holds the one action word and one question about the story. Otherwise one sentence with one
    action word and one question is enough.
15. Sound like a friendly teacher talking to a 7-year-old. Do not start with "Tell me". Do not stack several
    action words. You may vary between look, check, compare, notice and count; do not always use the same
    one.

Return JSON with one field, "tutor_text": the sentence(s) the child will hear.
```

S3.2 Viewpoint guidance (view-conditioned condition only; appended)

```
VR viewpoint guidance (v4, spatial wording only):
- The verified mathematical state is authoritative; viewpoint information is not evidence of correctness.
- Decide which region the learner still needs to work on, using arrangement_description, verified_errors and
    the story.
- If that region is not_visible and learner_has_seen_name is true: your reply MUST start with "Turn toward
    <region_name>." and then ask one question about the matching part of the story. Do not send the learner
    to a different, visible region instead.
- If that region is not_visible and learner_has_seen_name is false: describe its role using the story and
    start with "Turn to find the part of the story about <what it holds>." Do not use its name.
- If partially_visible: start with "Adjust your view toward <region_name>." only if the learner needs to see
    more of it; otherwise no turning words.
- If mostly_visible, or if the region the learner needs is visible: no turning words at all, even if some
    other region is out of view.
- If visibility is unknown or missing, do not claim the learner can see any particular region.
- Never claim an off-screen object is present or absent, reveal a hidden answer, or invent an object location.
- Never write a turning phrase in the same reply as the number of groups, parts or objects the story needs.
```

S3.3 Example per-request payload (view-conditioned; emoji fields omitted)

In the reference condition the field `vr_pov_context` is absent.


```
{
  "support_request": "HINT",
  "interaction": "EQUAL_GROUPS",
  "unknown_position": "TOTAL",
  "hidden_role": "TOTAL",
  "problem": {
    "story": "Ben was also given 2 trays of eggs to examine. If each tray has 5 eggs,",
    "question": "how many eggs did Ben examine in total?",
    "stated_quantities": {
      "number of groups": 2,
      "objects in each group": 5
    }
  },
  "object_language": {
    "object_label": "egg",
    "object_plural": "eggs",
    "group_label": "group"
  },
  "verified_arrangement": {
    "interaction": "EQUAL_GROUPS",
    "learner_group_count": 3,
    "learner_group_sizes": [
      6,
      5,
      5
    ],
    "undistributed_count": 0,
    "work_correct": false,
    "construction_status": "incorrect",
    "answer_entered": false,
    "answer_supported_by_work": true,
    "student_correct": false
  },
  "construction_status": "incorrect",
  "verified_errors": [
    "3 groups built, story has 2",
    "group 1 has 6, story puts 5 in each"
  ],
  "allowed_numbers": [
    0,
    1,
    2,
    3,
    5,
    6
  ],
  "latest_action": null,
  "learner_message": null,
  "recent_conversation": [],
  "vr_pov_context": {
```

```
    "regions": [
      {
        "visibility": "not_visible",
        "label_visible_now": false,
        "label_seen_this_attempt": true,
        "region_name": "Group 1",
        "learner_has_seen_name": true
      },
      {
        "visibility": "mostly_visible",
        "label_visible_now": true,
        "label_seen_this_attempt": true,
        "region_name": "Group 2",
        "learner_has_seen_name": true
      },
      {
        "visibility": "mostly_visible",
        "label_visible_now": true,
        "label_seen_this_attempt": true,
        "region_name": "Group 3",
        "learner_has_seen_name": true
      }
    ]
  }
}
```


S3.4 Output schema and generation settings

Strict JSON schema: an object with the single required string property `tutor_text` and no additional properties. Settings (both conditions): model `gpt-6.1-sol` via the OpenAI Responses API, low reasoning effort, 400-token output limit, 20 s timeout, no retries. **Field-name note.** Rule 8 (both conditions) and the viewpoint guidance (view-conditioned condition) refer to an `arrangement_description`; in the frozen prompt version used here that field was not part of the payload, and the arrangement was supplied as `verified_arrangement` together with `verified_errors`. The instruction text is reproduced unchanged. Rule 14 refers to a "turn phrase"; this wording is present in both conditions, but only the view-conditioned condition receives a viewpoint to which it could apply.

S4. Evaluation set

Twenty scripted non-correct states from 20 distinct problems in the bank of 88 (10 incomplete, 10 incorrect). The states were also used during prompt development and were not held out. Each state was run under five views, two repeats, and two conditions (400 hints). Forty correct-state confirmations did not call the model and are excluded.

Task type	State (question id / scenario)	Status	Primary relevant	All discrepant regions	Regions
EQUAL_GROUPS	0377/extra_group	incorrect	group__7	group__7	7
EQUAL_GROUPS	0400/two_errors	incorrect	group__1	group__1, group__5	5
EQUAL_GROUPS	0409/two_errors	incorrect	group__1	group__1, group__3	3
EQUAL_GROUPS	1771/two_errors	incorrect	group__1	group__1, group__4	4
EQUAL_SHARING	0464/dealing_out	incomplete	recipient__1	recipient__1, recipient__2, recipient__3, recipient__4	5
EQUAL_SHARING	0483/two_shares_too_big	incorrect	recipient__1	recipient__1, recipient__2, recipient__3	6
EQUAL_SHARING	0577/one_share_too_big	incorrect	recipient__1	recipient__1, recipient__2	3
EQUAL_SHARING	1582/nothing_shared	incomplete	recipient__1	recipient__1, recipient__2	3
JOIN	0001/parts_incomplete	incomplete	part__b	part__b	4
JOIN	0004/totals_incomplete	incomplete	part__a	part__a, part__b	4
JOIN	0029/parts_error_while_unfinished	incorrect	part__a	part__a, part__b	4
JOIN	0077/parts_two_errors	incorrect	part__a	part__a, part__b	4
PART_TO_PART_RATIO	0850/nothing_placed	incomplete	part__a	part__a, part__b	4
PART_TO_PART_RATIO	0851/first_part_done	incomplete	part__b	part__b	4
PART_TO_PART_RATIO	0854/first_part_done	incomplete	part__b	part__b	4
PART_TO_PART_RATIO	1146/nothing_placed	incomplete	part__a	part__a, part__b	4
REMOVE	0025/one_too_many_removed	incorrect	removed	removed, start	2
REMOVE	1724/partly_removed	incomplete	removed	removed, start	2
REMOVE	1737/one_too_many_removed	incorrect	removed	removed, start	2
REMOVE	1742/nothing_removed	incomplete	removed	removed, start	2

S5. Annotation criteria

One author annotated all 400 hints, shuffled, with the prompt condition hidden and the recorded visibility snapshot available (blinding was partial because orientation wording can reveal the condition). Criteria, as given to the annotator:

- **Specific spatial reference.** A phrase directing the student to a particular workspace region using a name, an object noun used spatially, a description, a pointing expression, or orientation. Story-only and whole-construction phrases are not specific references. A story noun is spatial only when used to identify a VR region; a region is not inferred from story events alone.
- **Category.** 1 = uniquely resolved region or defined set; 2 = multiple plausible regions; 3 = no specific workspace reference; 4a = specific reference unresolvable from wording; 4b = cannot decide because scene information or criteria are insufficient.
- **Resolved referent(s), role.** Exact region name(s) or well-defined set; role is target, anchor, prospective (region the task requires creating; not "nonexistent"), mixed, other, or not applicable.
- **Mathematical target.** Explicitly identified region(s) separated by "+"; *construction* for whole-construction issues; *none* for story-only hints.
- **Admissibility.** Yes if target and anchors suit the task (given the verified state and available comparison regions); interchangeable regions are equally admissible.
- **Visibility consistency.** Judged against the recorded view. Naming or turning toward a hidden region is allowed; claiming a hidden region is immediately visible is not. Direction errors are not inferred without recorded geometry. Deixis on a partially visible region is not a contradiction; presupposing its full contents are visible is.
- **Hard grounding failure.** Ambiguous (category 2) or unresolvable (4a) reference; nonexistent referent; inadmissible target or anchor; scene- or visibility-dependent claim that contradicts the snapshot.
- **Orientation phrase.** Yes for explicit instructions to turn, look around, or orient toward a region.

Metrics. *Coverage* = hints with category 1, 2, 4a, or 4b. *Grounded success* = category-1 hints that are admissible, visibility-consistent, and without hard failure, over evaluable (1, 2, 4a) hints. *Grounded yield* = the same numerator over all hints except category 4b (199 of 200 view-conditioned hints). Category 4b is excluded from the denominators of grounded success and grounded yield, not from coverage.

Mathematical focus (RQ2). Hints for the same state are compared in pairs matched by generation index across views (relevant-only vs. partial, partial vs. hidden-seen, relevant-only vs. hidden-seen; 120 pairs) and within each view (60 pairs). Pairs whose annotated targets are identical are counted as the same focus. For the 76 pairs with differing targets, the annotator assigned one class (66 view-conditioned, 10 reference): *same* (same question about the same kind of quantity, including swaps between regions with identical content), *different* (a different quantity or part of the problem is addressed), *region swap* (the same question about a different region with different content), or *uncertain*. Region swaps and uncertain cases are not counted as focus changes; counting them gives an upper bound. These classifications were not independently checked. Pair-level data: `evaluation/target_pair_classification.csv`; `evaluation/reproduce_results.py` recomputes the tables and intervals from the two CSV files. Where the two repeats of a view are identical, their order is a labeling convention; every published pair is index-matched.

S6. Example hints

All hints are unedited model outputs for one state (equal groups, two errors; "Ben was also given 2 trays of eggs…"; learner groups of sizes 6, 5, 5). Two view-conditioned generations and the first reference generation are shown per view. The relevant region is Group 1.

View	View-conditioned, gen. 1	View-conditioned, gen. 2	Reference, gen. 1
all-visible	Compare your groups with Ben's trays in the story–does the number of groups match?	Check the eggs in Group 1–does that group match what the story says each tray holds?	Compare your groups with the trays in Ben's story–does the number of groups match?
relevant-only	Check the eggs in Group 1–does that group match what the story says is in each tray?	Check Group 1: does it have the eggs each tray should hold in Ben's story?	Compare your number of groups with the number of trays in Ben's story–do they match?
partial	Check how many trays Ben was given in the story–does your number of groups match?	Compare your number of groups with the number of trays in Ben's story–do they match?	Check your groups–does the number of groups match Ben's trays in the story?
hidden-seen	Turn toward Group 1. Check the eggs there–does this match what the story says each tray holds?	Turn toward Group 1. Check the eggs there–does that match how many eggs the story says are in each tray?	Compare your number of groups with the number of trays in Ben's story–do they match?
hidden-unseen	Check how many trays Ben was given in the story–does your number of groups match?	Check your groups: does the number of groups match the number of trays in Ben's story?	Compare how many groups are on your desk with how many trays Ben was given–do they match?

Under partial and hidden-seen views the view-conditioned hints address different issues (number of groups vs. eggs in Group 1), the pattern behind the focus changes reported in the paper. The four view-conditioned grounding failures are listed in `examples/example_hints.json`.

S7. Additional results

Per task type (view-conditioned hints, 40 per task type).

Task type	Coverage	Grounded success	"Turn toward" (hidden-seen)	"Turn to find…" (hidden-unseen)
`JOIN`	30/40	29/30	6/8	5/8
`REMOVE`	36/40	36/36	8/8	8/8
`EQUAL_GROUPS`	16/40	15/15	8/8	1/8
`EQUAL_SHARING`	28/40	25/28	5/8	3/8
`PART_TO_PART_RATIO`	33/40	33/33	8/8	8/8

Totals: coverage 143/200, grounded success 138/142, required opening phrase 35/40 (hidden-seen) and 25/40 (hidden-unseen). The hidden-unseen opening was rarely produced for equal groups (1/8) and equal sharing (3/8). Coverage is lowest for equal groups, where many hints address the whole construction (e.g., the number of groups) and name no region.

Generation and protection. All 400 requests succeeded; for all 400 the final text equals the generated text (no blocked hints, no fallbacks). The longest hint has 29 words; the largest output used 212 tokens (179 reasoning). Median API latency was about 2.8 s (view-conditioned) and 3.0 s (reference).